

Listing 9-1: Sending the “speak” message to objects instantiated from different classes

Each class has a `speak()` method, but the content of each method is different ❶ ❷ ❸. Each class does whatever it needs to do in its version of this method; the method name is the same, but it has different implementations.

To make things easy to deal with, we put all the pet objects into a list ❹. To make them all speak, we then loop through all the objects and send the same message by calling a method with the exact same name in each object ❺, without worrying about the type of the object.

Example Using Pygame Shapes

Next, we’ll look at a demonstration of polymorphism using pygame. In Chapter 5 we used pygame to draw primitive shapes such as rectangles, circles, polygons, ellipses, and lines. Here we’ll build a demonstration program that will randomly create and draw different shapes in a window. The user can then click on any shape, and the program will report the type and area of the shape that was clicked. Because the shapes are randomly created, each time the program runs, the size, location, number, and position of the shapes will be different. [Figure 9-1](#) shows some sample output from the demonstration program.